TheLook eCommerce Analytics: Project Report

Excel only: Power Query, Power Pivot (Data Model), DAX, Forecast Sheet. Training project on the public TheLook dataset (7 CSV tables).

1. Question and result

TheLook struggles with high cart abandonment and rising product returns. The goal was to find where and why this happens in the customer journey. **Result:** the company is profitable at gross margin level (51.89%), but 58% of carts do not convert, 29% of delivered units are returned, and inventory covers about 22 months of demand. Both customer problems are **uniform across every segment tested**, which points to store-wide causes rather than a specific channel, product, or day.

2. Data and model

Table	Rows	Grain / role
order_items	181,759	One item sold. Fact table
events	2,431,963	One site event (681,759 sessions). Fact table
orders	125,226	One order
users	100,000	One customer
products	29,120	One product
inventory_items	490,705	One physical item (308,946 in stock)
distribution_centers	10	One center
Calendar	-	Date table (Power Pivot)

Preparation (Power Query). Columns were chosen with a rule: keep keys, numbers to aggregate, dimensions to slice by, and dates. Heavy high-cardinality text (email, street address, SKU, IP, URI) and columns duplicated across tables (product attributes in inventory_items, dates in orders) were removed, because the Data Model compresses by distinct values and duplicate columns create two sources of truth. Nulls were kept where they carry meaning (returned_at empty = not returned, sold_at empty = in stock, user_id empty = anonymous visitor). All tables were loaded as Connection Only into the Data Model.

Model. Star schema with two fact tables sharing dimensions. All relationships are one-to-many. Users sits in the middle between events and order_items to avoid many-to-many. inventory_items is not related to distribution_centers (it would create two active paths), so center names are looked up with LOOKUPVALUE.

3. Findings

Customer journey (sessions, not events).

Stage	Sessions	Rate
All sessions	681,759	
With a cart event	432,146	63.39% of sessions
With a purchase event	181,759	42.06% of cart sessions
Conversion / cart abandonment		26.66% / 57.94%

Channels and browsers convert the same. Facebook 26.95%, Email 26.67%, Adwords 26.60%, Organic 26.60%, YouTube 26.51%. Browsers range from 26.10% (IE) to 26.79% (Chrome). Weekday conversion ranges from 23.75% to 24.36%, and Facebook on weekends is not different from other days. The dataset has no advertising cost, so ROI cannot be calculated.

Revenue. Recorded sales are $10.83M, but 14.8% is cancelled and 10.1% is returned. Net revenue is **$8.13M**, gross profit $4.22M, gross margin 51.89% (no operating or ad costs in the data). 62.26% of buyers ordered once and 37.74% came back.

Returns. The return rate depends on the base: 10.03% on all items, 11.79% excluding cancelled items, and **28.56% on delivered items** (18,232 of 63,841), which is the correct base because an undelivered item cannot be returned. Category rates range from 26.61% to 30.97%, departments 28.36% and 28.76%, distribution centers 27.79% to 30.18%, and brands with 300+ delivered units 31.04% to 33.33% (top five). Variation is within what chance produces across so many comparisons, so no category, brand, or center stands out. New Orleans is slightly higher but the effect is about 100 returns over six years.

Forecast and inventory. An Excel ETS model on 1,826 daily observations (2019 to 2023) forecasts about 461 units per day for Jan to Mar 2024 (95% upper bound 501). In a 10-day test (Jan 1 to 10, 2024) it came out about 11% below actual and actual demand exceeded the upper bound, so it underestimates growth. In-stock inventory is 308,946 units: about 670 days (22 months) of coverage, and about 11 months even if demand doubles. Coverage is similar across all centers and categories. The risk is excess stock, not stockouts.

4. Corrections made during the analysis

- Return rate: 10.03% (all items) became 11.79% (cancelled removed) and finally 28.56% (delivered base).
- Total sales included cancelled and returned items (25% of the total). Net revenue is used instead.
- The first 1,000 rows suggested user_id was empty. Full profiling showed 46% empty (anonymous visitors).
- Daily and yearly session totals do not add to the distinct total because a session crossing a period boundary counts in both. The distinct count is the correct total.
- January 2024 looked complete, but orders end on Jan 21 and the last days show an unusual spike and fall, so it was excluded from forecast training.

5. Recommendations

#	Observation	Reason	Action
1	58% of cart sessions end without purchase	Same in every channel, browser, and weekday, so likely store-wide	Review checkout (shipping, final price, steps) and run an A/B test
2	Paid channels convert like organic	Conversion alone does not show financial return	Collect cost per channel before moving budget
3	28.56% of delivered units are returned (about $1.1M)	Uniform across categories, brands, departments	Improve size guides and descriptions, record return reasons, pilot on one category
4	About 22 months of inventory at forecast demand	Excess stock across all centers	Reduce new purchasing, retrain the forecast monthly, use the upper bound as the safe case
5	46% of events lack user_id; purchase events appear to be per item	Limits customer-level analysis	Log purchases per order and link visitors to accounts

6. Limitations

- No advertising or operating cost: margin is gross, and channel ROI cannot be assessed.
- Synthetic dataset: real customer behavior may differ, and the size of the numbers is not a market benchmark.
- Purchase sessions equal the item count, so order-level conversion is lower than 26.66% (inferred, not confirmed).
- Net revenue includes shipped and processing items that may still be cancelled or returned, so 28.56% is a minimum.
- Returned items may re-enter stock, so available inventory may be higher (assumption).
- The forecast was tested on 10 days only and tends to underestimate demand.
- Retention is approximated by repeat purchase rate (37.74%), which is not time-based.